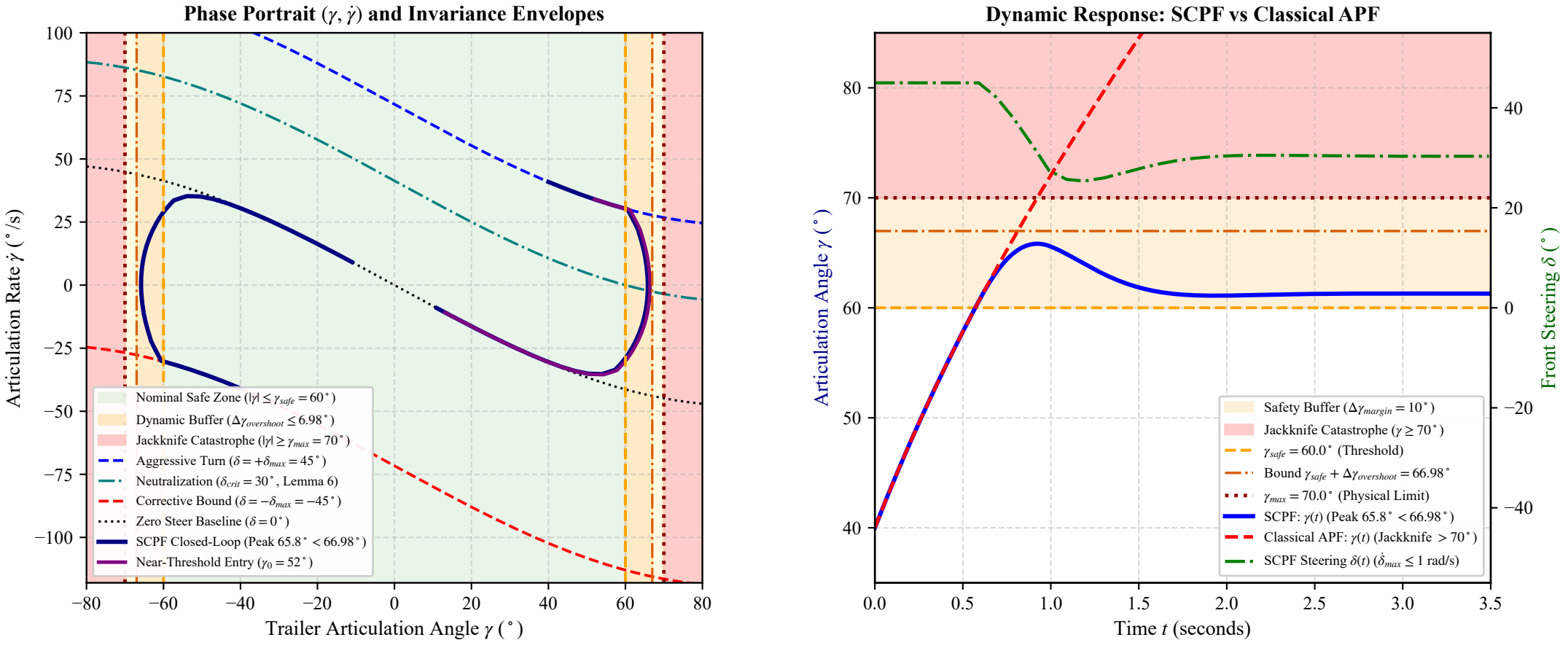

Phase Portrait (γ, $\dot{\gamma}$) and Invariance Envelopes
Articulation Rate $\dot{\gamma}$ (°/s)
Trailer Articulation Angle γ (°)
Nominal Safe Zone ($|\gamma| \leq \gamma_{safe} = 60°$)
Dynamic Buffer ($\Delta\gamma_{overshoot} \leq 6.98°$)
Jackknife Catastrophe ($|\gamma| \geq \gamma_{max} = 70°$)
Aggressive Turn ($\delta = +\delta_{max} = 45°$)
Neutralization ($\delta_{crit} = 30°$, Lemma 6)
Corrective Bound ($\delta = -\delta_{max} = -45°$)
Zero Steer Baseline ($\delta = 0°$)
SCPF Closed-Loop (Peak 65.8° < 66.98°)
Near-Threshold Entry ($\gamma_0 = 52°$)
Dynamic Response: SCPF vs Classical APF
Articulation Angle γ (°)
Front Steering δ (°)
Time t (seconds)
Safety Buffer ($\Delta\gamma_{margin} = 10°$)
Jackknife Catastrophe ($\gamma \geq 70°$)
$\gamma_{safe} = 60.0°$ (Threshold)
Bound $\gamma_{safe} + \Delta\gamma_{overshoot} = 66.98°$
$\gamma_{max} = 70.0°$ (Physical Limit)
SCPF: $\gamma(t)$ (Peak 65.8° < 66.98°)
Classical APF: $\gamma(t)$ (Jackknife > 70°)
SCPF Steering $\delta(t)$ ($\dot{\delta}_{max} \leq 1$ rad/s)